Retry

Season: 4 · Enterprise · Category: Enterprise · Level: Intermediate

Intent

- Re-execute transient failures with policy.

Problem

- Without this pattern, code tends toward: New things up with conditionals and duplicated construction.

Real-world Analogy

- Redialing a busy phone line.

When to Use

- Microservices / modular monoliths.
- The design force named in Intent shows up more than once.

When Not to Use

- Greenfield CRUD without distribution.
- A simpler language feature (record, sealed switch, function value) already solves it.

Pattern Structure

- See mermaid diagram in explanation.md

Classic Implementation

```java
package com.buildwithwaqar.patterns.enterprise.retry;

import java.util.concurrent.Callable;
import java.util.function.Predicate;

public final class Retry {
    private final int maxAttempts;
    private final Predicate<Exception> retryOn;

    public Retry(int maxAttempts, Predicate<Exception> retryOn) {
        this.maxAttempts = maxAttempts;
        this.retryOn = retryOn;
    }

    public <T> T execute(Callable<T> action) throws Exception {
        Exception last = null;
        for (int i = 1; i <= maxAttempts; i++) {
            try {
                return action.call();
            } catch (Exception e) {
                last = e;
                if (i == maxAttempts || !retryOn.test(e)) throw e;
            }
        }
        throw last;
    }
}
```

Modern Java 25 / Functional

- Sealed events + pattern matching for projections.

Worked Example

```
See the JUnit test beside the implementation.
Run: mvn -Dtest=*Test test
Focus package matching: enterprise/retry/
```

Pitfalls

- Keep demos minimal — production needs idempotency and observability.
- Saga compensations must be careful.

Interview Q&A

- Q: Production concern for Retry? A: Idempotency, observability, and failure modes.
- Q: How would you test this? A: Prefer injecting interfaces or interpreters; keep demos deterministic (fixed clocks, canned inputs).

Key Takeaways

- Re-execute transient failures with policy.
- Prefer Java 25 sealed types / records / virtual threads where they simplify the classic shape.
- Ship working tests with every episode.

Video Script Highlights

- [0:00] Hook — Open with the analogy: Redialing a busy phone line. [0:40] Intent — State the design force in one sentence: Re-execute tran...